AITOGY-GNSS-BASE-ESP-UM980

H ng d n kh ch h ng | Firmware V2.3.5 | C p nh t 25/10/2026

1. Ph m vi v nguy n t c v n h nh

B AITOGY-BASE-ESP nh n d li u t receiver GNSS, chuy n ti p RTCM l n caster b ng 4G ho c Wi-Fi v nh n i u khi n t h th ng CORS. Anten base ph i ng y n khi ph t c i ch nh. Sau khi di chuy n anten, c u h nh l i t a base tr c khi cho rover s d ng.

T i li u m t firmware **V2.3.5**, c p nh t **25/10/2026**. K t qu bi n d ch v ki m tra ph n m m kh ng thay th nghi m thu ngo i th€c a. Kh ng cam k t th•i gian kh , i ng c nh hay ch nh x c o khi ch a ki m tra receiver, anten, m ng c i ch nh v h t a .

Ch	Khi s d ng	i u ki n ho n th nh
FIXED_LLA	ƒ„ c… t a i m t anten ng tin c y.	G i c u h nh fixed, ki m tra t a receiver v lu†ng RTCM.
SURVEY_IN	C n receiver t€ kh o s t v tr ng y n.	Receiver k t th‡c kh o s t; g i l nh th nh c ng kh ng c… ngh‚a kh o s t „ xong.
AUTO_CORS	C… t i kho n rover nh n c i ch nh t CORS.	Chuy n rover, nh n c i ch nh, l y m‰u RTK_FIXED, t l i base v ki m tra ph t RTCM.

ƒ cao ph i dŠng ‡ng quy c c‹a t a cung c p. V i c u h nh fixed theo LLA, s d ng cao ellipsoid. Kh ng thay tr€c ti p b ng cao th‹y chu n ho c b• qua chi u cao anten/ i m quy chi u.

2. K t n i v chu n b

ng t n hi u TDM240x	K t n i
GPIO18 / RX GNSS c‹a ESP32	Nh n t TX c‹a receiver GNSS.
GPIO19 / TX GNSS c‹a ESP32	G i v o RX c‹a receiver GNSS.
GND	Chung mass gi a ESP32, receiver v b ngu†n theo thi t k b thi t b .
GPIO16 / RX modem	Nh n t TX modem.
GPIO17 / TX modem	G i v o RX modem.
Serial console	115200 baud; g i m t dŽng l nh v k t th‡c b ng LF.
UART GNSS	M c nh 115200 baud, 8N1.

ƒ i chi u nh„n c•ng v s• † c‹a ‡ng board tr c khi u d‘y. Kh ng suy ra ch‘n c’m hay m c ngu†n t h nh nh c‹a board kh c. G’n anten GNSS ‡ng c•ng, t n•i tho ng tr•i, c nh c• kh ; ki m tra SIM/APN v anten 4G tr c khi ch y auto base.

B n Unicore dŠng UM980 ho c UM982 v i b l nh ASCII t •ng th ch. ƒ •ng UART firmware n i COM2 receiver; kh ng m c nh coi c•ng USB/Type-C tr n m i board l COM2. C n x c nh n s• † ‡ng module. Vi c •i c•ng receiver ph i do k“ thu t c u h nh †ng b v i d‘y n i.

3. C u h nh m c nh v th ng tin c n b n giao

Th ng s	Gi tr / ngh a trong firmware
K t n i m ng	4G; c… th chuy n Wi-Fi b ng ATG.
APN	v-internet; thay theo SIM th€c t .
MQTT broker m c nh	45.117.179.134:1883; ph i kh p h th ng CORS ang tri n khai.
Backend WebSocket	base.aitogy.com:8000, giao th c ws; c n endpoint thi t b th€c s€ truy c p ”c.
Caster d€ phŽng trong c u h nh m c nh	aitogy.com.vn:2101, mountpoint /test; kh ng ph i th ng tin c p ph t cho m i kh ch h ng.
K nh qu n l• ch nh	Topic theo serial esp_... do thi t b t o; qu n tr vi n c p quy n tr n CORS.
Tr ng th i backend	Chu k– m c nh 5 gi‘y; k t n i m ng c… th l m ch m vi c g i.
Health ki u c–	Chu k– m c nh 30 gi‘y.

ƒ•n v tri n khai b n giao: serial, t n tr m, lo i GNSS, caster/mountpoint ph t, t i kho n c i ch nh u v o n u dŠng AUTO_CORS, APN ho c Wi-Fi v ng •i qu n l• tr n CORS. M t khŒu/token do qu n tr vi n c p ri ng, kh ng l y t v d trong t i li u.

4. a thi t b l n CORS

1. C p ngu†n •n nh, m , console 115200 baud v ghi l i serial esp_..., phi n b n firmware, lo i chip, baud nh n di n ”c.
2. C u h nh m ng, MQTT v backend theo tri n khai. C c chu˜i trong d u <...> l gi tr c n thay, kh ng nh p nguy n d u ngo c.
3. L u th ng tin tr c, g i l nh ch n m ng sau cŠng v l nh … kh , i ng l i thi t b .
4. Tr n CORS, qu n tr vi n nh n/c p thi t b , t t n, tri n khai license v ch n ‡ng lo i chip.
5. Tri n khai d ch v caster: server 1, server 2 n u c n, mountpoint, t i kho n ph t v ch lu n ph t/theo y u c u.
6. Ch n FIXED_LLA, SURVEY_IN ho c AUTO_CORS. Theo d™i k t qu l nh v ti n , sau … ki m tra lu†ng RTCM b ng rover c l p.

V d c u h nh 4G:

```
ATG ESP SET 4G APN <apn>
ATG MQTT SET SERVER <mqtt_host>
ATG MQTT SET PORT 1883
ATG MQTT SET USER <mqtt_user>
ATG MQTT SET PASS <mqtt_password>
ATG ESP SET BACKEND HOST <backend_host>
ATG ESP SET BACKEND PORT 8000
ATG ESP SET CONNECTION 4G
```

V d •i sang Wi-Fi:

```
ATG ESP SET WIFI SSID <ssid>
ATG ESP SET WIFI PASS <wifi_password>
ATG ESP SET CONNECTION WIFI
```

Console ATG t ch tham s theo d u c ch, kh ng h˜ tr" chu˜i c… kho ng tr'ng ho c gi tr r˜ng. N u SSID/m t khŒu c n d ng n y, li n h k" thu t; kh ng dŠng d u nh y gi l p chu˜i.

Thi t b ch a "c c p quy n ho c ch a c… license v‰n c… th b o tr ng th i nh ng kh ng ‹ i u ki n ph t RTCM. Tr ng th i online c‹a k nh qu n l• c–ng ch a ch ng minh rover nh n "c c i ch nh.

5. Ch n ch base v x c nh n k t qu

5.1 T a c nh

Nh p vˆ /kinh d ng th p ph'n v cao ellipsoid t nh b ng mšt. Ki m tra ‡ng i m quy chi u anten v h t a . Sau khi c u h nh, i chi u t a c t receiver, s€ hi n di n c‹a b n tin t a base v t a rover t i i m ki m tra.

5.2 Survey-in

ƒ t anten ng y n, nh p th•i gian kh o s t theo gi'y. Nh nh Unicore hi n sinh mode base time <seconds>; tham s accuracy c‹a giao di n ATG v‰n b't bu c v c‡ ph p nh ng kh ng "c a v o l nh survey/fixed ASCII n y.

Survey-in kh ng t€ t o ra t a tuy t i t ch nh x c kh o s t ch› v ch n ng œng nh•. Khi c n t a g'n v i h quy chi u c th , dŠng i m „ bi t ho c AUTO_CORS v th€c hi n ki m tra c l p.

5.3 Auto base t€ CORS

Firmware chuy n receiver sang rover, nh n RTCM t mountpoint c i ch nh v ch› ch p nh n GGA m i v i tr ng th i **RTK_FIXED**. RTK_FLOAT kh ng "c dŠng l m m‰u auto base.

Lu†ng m c nh c n 5 l n ki m tra FIX li n ti p theo nh p kho ng 1 gi'y, sau … l y t i a 60 m‰u trong c a s• 60 gi'y. C n t nh t 15 m‰u h"p l ti p t c; kh ng m c nh coi k t qu lu n c… ‹ 60 m‰u. Thi t b t l i t a base r†i kh i ph c d ch v ph t „ c u h nh.

Th•i gian ch• FIX m c nh 3600 gi'y; qu n tr vi n c… th c u h nh theo nghi p v . Khi m t FIX, m t k t n i, thi u m‰u ho c c… l nh d ng, xem l• do trong ti n workflow. Vi c kh i ph c k t n i caster sau th t b i kh ng t€ ch ng minh receiver „ tr , v c u h nh base mong mu n.

Ph n chuy n VN2000 hi n dŠng tham s t ch h"p v phšp chi u theo l€a ch n t›nh/kinh tuy n. Firmware kh ng gi i m„ c c b n tin datum RTCM 1021/1023/1025. Ph i i chi u m c ki m tra th€c a; kh ng coi nh„n VN2000 l ch ng nh n ‡ng h t a .

6. Theo d•i tr ng th i v lu , ng d ƒ li u

Hi„n th /tr ng	C ch hi„u
awaiting_license / unprovisioned	Ch a ‹ i u ki n c p quy n/license ph t RTCM.
locked	Thi t b b kh…a to n b ; li n h qu n tr vi n.
sleep	ƒang ch• k ch ho t lu†ng theo y u c u; ch a ch'c l m t m ng.
online	K nh qu n l• ho t ng; c n ki m tra th m t ng server NTRIP.
auto_setup_* / reference_check*	Workflow ang th€c hi n m t b c c u h nh ho c ki m tra.
last_command_result	K t qu x l• l nh; l nh kh , i ng workflow th nh c ng ch› ngh^a „ nh n vi c.
auto_base_progress	Theo d™i phase, s m‰u v l• do completed/failed/stopped c‹a workflow.
server1/2_connected v server1/2_bps	K t n i v s byte/gi'y g i t i t ng caster.
rtcm_input_bps	S byte RTCM/gi'y "c …ng khung , UART; kh ng ph i ch ng nh n CRC h"p l .
GGA/fix/satellites	Ch t l "ng nghi m GNSS v d li u v tr ang nh n.

V d c u tr‡c health ki u c– (gi tr minh h a):

```
{
  "uptime_s": 120,
  "free_heap_bytes": 120000,
  "connected_via": "GSM",
  "rssi_dbm": 20,
  "mqtt_ok": true,
  "ntrip_ok": true,
  "gnss_data_ok": true,
  "rtcm_decoder": false,
  "rtcm_types": {}
}
```

connected_via dŠng GSM ho c WiFi. gnss_data_ok ph n nh „ nh n khung RTCM g n 'y, trong ng œng 5 gi'y c‹a t c v health. rtcm_types r˜ng v ESP32 kh ng m t ng lo i b n tin. rssi_dbm l t n tr •ng c–: Wi-Fi dŠng RSSI, cŽn 4G ang gi gi tr ch t l "ng t n hi u th do modem tr v , ch a t€ chuy n sang dBm. Kh ng dŠng gi tr … nh s dBm 4G „ hi u chu Œn.

H th ng nh n NMEA GGA/GSA/GSV/GST theo d™i GNSS. ESP32 kh ng m c nh t o m t b n sao to n b RTCM l n MQTT. C c tr •ng c… t n valid_packets hi n l b m …ng khung, kh ng c… ki m tra CRC RTCM. hacc c… th l y t GST ho c fallback HDOP; kh ng coi m i gi tr hacc u l ch nh x c mšt „ x c nh n.

7. Danh m c l nh ATG

ATG dŠng c u h nh k t n i v h˜ tr" k" thu t. •u ti n CORS c u h nh base v c k t qu l nh theo t ng thi t b . T kh…a dŠng ‡ng ch hoa; m˜i dŽng k t th‡c b ng LF. L nh SET th •ng l u c u h nh, c n kh , i ng l i k t n i c l i gi tr m i.

7.1 ESP v m ng

C… ph p	T c d ng
ATG ESP RESTART	Kh , i ng l i ngay.
ATG ESP SET CONNECTION 4G	L u ch n 4G v kh , i ng l i.
ATG ESP SET CONNECTION WIFI	L u ch n Wi-Fi v kh , i ng l i.
ATG ESP SET WIFI SSID <ssid>	L u SSID kh ng c… kho ng tr ' ng.
ATG ESP SET WIFI PASS <password>	L u m t khŒu kh ng c… kho ng tr ' ng.
ATG ESP SET 4G APN <apn>	L u APN.
ATG ESP SET 4G USER <user>	L u t n žng nh p APN khi nh m ng y u c u.
ATG ESP SET 4G PASS <password>	L u m t khŒu APN khi nh m ng y u c u.
ATG ESP SET BACKEND HOST <host>	L u hostname/IP c‹a endpoint WebSocket, kh ng kŸm scheme ho c •ng d‰n.
ATG ESP SET BACKEND PORT <port>	L u c•ng WebSocket; ch p nh n 1-65535.

7.2 MQTT v NTRIP ph t

C… ph p	T c d ng
ATG MQTT SET SERVER <host>	L u broker MQTT.
ATG MQTT SET PORT <port>	L u c•ng MQTT; nh p s h ” p l 1-65535.
ATG MQTT SET USER <user>	L u t n žng nh p broker.
ATG MQTT SET PASS <password>	L u m t khŒu broker.
ATG MQTT SET PUBTPCHEALTH <topic>	L u topic health ki u c–.
ATG MQTT SET SUBTPCCMD <topic>	L u topic subscribe ATG ki u c–; thay topic c n k“ thu t ki m tra callback th€c nh n.
ATG NTRIP SET CSTRADDR <host>	L u caster m c nh/d€ phŽng.
ATG NTRIP SET CSTRPORT <port>	L u c•ng caster; nh p s h ” p l 1-65535.
ATG NTRIP SET MNTPNT <mountpoint>	L u mountpoint m c nh.
ATG NTRIP SET CSTRAUTH <password>	L u m t khŒu source nguy n vžn; kh ng nh p base64 c‹a username:password.

D ch v „ c p t CORS c… th u ti n h•n c u h nh NTRIP m c nh l u b ng ATG. Mu n thay t i kho n/mountpoint ang v n h nh, ch›nh d ch v tr n CORS r†i tri n khai l i.

7.3 GNSS v t l i c u h nh

C… ph p	T c d ng
ATG GNSS ROVER	Chuy n receiver sang rover ch€n o n; kh ng t€ žng nh p t i kho n rover CORS.
ATG GNSS BASE SURVEY_IN <seconds> <accuracy_m>	G i batch survey-in theo lo i chip; m c nh ch› xu t RTCM, kh ng xu t NMEA diagnostic.
ATG GNSS BASE SURVEY_IN <seconds> <accuracy_m> W_DIAG	Survey-in v xu t c RTCM l‰n c c b n tin NMEA diagnostic; W_DIAG kh ng ph‘n bi t hoa th •ng.
ATG GNSS BASE FIXED <lat> <lon> <alt> <accuracy_m>	G i t a LLA c nh; cao t nh b ng mšt ellipsoid.
ATG GNSS BASE <message> <port> ON	B t b n tin RTCM tr n c•ng ch› nh.
ATG GNSS BASE <message> <port> OFF	T’t b n tin RTCM tr n c•ng ch› nh.
ATG GNSS SET CHIP UNICORE	L u l€a ch n chip v kh , i ng l i; boot v‰n th€c hi n nh n di n ph n c ng.
ATG CONFIG RESET	ƒ t c• n p l i m c nh v kh , i ng l i; c… th l m •i m ng v c u h nh GNSS.

C•ng dŠng COM1/COM2/COM3, •ng n i ESP32 m c nh COM2. V d ch€n o n output: ATG GNSS BASE 1019 COM2 ON. •u ti n thao t c base qua CORS v •ng ATG c– ch a ki m tra OK ri ng t ng l nh.

L nh ATG ki u c– ch‹ y u tr log Serial, kh ng c… k t qu JSON ri ng theo command_id nh l nh CORS. Kh ng coi dŽng „ g i¡ l receiver „ ch p nh n c u h nh. Reset c u h nh c c b kh ng thay th thao t c gœ thi t b /thu h†i quy n tr n CORS; qu n tr vi n ph i x l• tr ng th i c p ph t khi c n.

8. RTCM, ephemeris v c u h nh receiver

8.1 B n tin ph t

Nh†m	B n tin
T a /nh n d ng base	1006, 1033.
GPS	MSM4 1074 / MSM7 1077; ephemeris 1019.
GLONASS	MSM4 1084 / MSM7 1087; ephemeris 1020.
Galileo	MSM4 1094 / MSM7 1097; ephemeris 1045.
BeiDou	MSM4 1124 / MSM7 1127; ephemeris 1042.
QZSS	MSM4 1114 / MSM7 1117; ephemeris 1044.

Fixed/survey ch n nh…m theo h v tinh v m c MSM trong c u h nh; m c nh GPS/GLO/GAL/BDS b t, QZSS t’t. Ephemeris c‹a h b t kh ng b l c b• khi ch n MSM4 ho c MSM7. Bootstrap UM980 b t ri ng ‹ 1019/1020/1042/1044/1045 tr n COM2, chu k– y u c u 1 gi‘y, tr c saveconfig. B c u h nh LTE c… c hai m c MSM v c c ephemeris n y. Kh ng †ng nh t bootstrap v i m i c u h nh ”c c p v sau.

8.2 Ki„m tra tr‡c ti p Unicore

K t n i terminal v i ‡ng c•ng c‹a receiver, dŠng baud ang c u h nh v k t th‡c l nh b ng CRLF. ƒ'y l l nh g i tr€c ti p v o receiver, kh ng th m ti n t ATG.

L nh ASCII tr‡c ti p	Ch^c n‰ng
version	Truy v n model/phi n b n receiver.
mode	Truy v n ch ang t.
log	Truy v n output log ang c u h nh.
unlogall	Y u c u ng ng c c output log tr c khi c u h nh l i.
config com2 115200	ƒ t baud cho COM2; sau thay •i c n n i ‡ng baud m i.
mode rover	Chuy n sang rover; cŽn c n c i ch nh u v o c… RTK.
mode base time 300	Y u c u survey-in t i thi u 300 gi'y.
mode base 10.7500000000 106.5000000000 12.5000	V d fixed LLA; thay b ng t a th t c‹a anten, kh ng s d ng nguy n gi tr m‰u.
gpgga com2 1 / gpgst com2 1	Y u c u output NMEA ph c v theo d™i v tr /ch t l ”ng.
rtcm1074 com2 1	Y u c u output GPS MSM4 tr n COM2; ch n b b n tin phŠ h”p rover.
saveconfig	L u c u h nh receiver cho l n kh , i ng sau.
FRESET	Kh i ph c m c nh receiver; l m m t c u h nh „ t v c… th thay baud.

Kh ng g i to n b b ng nh m t batch m‰u. Ch n ‡ng l nh cho b c c u h nh, ch• ph n h†i v ki m tra tr ng th i; kh ng reset receiver ang khai th c ch› th l nh.

V d y u c u output ephemeris tr n COM2:

```
rtcm1019 com2 1
rtcm1020 com2 1
rtcm1042 com2 1
rtcm1044 com2 1
rtcm1045 com2 1
saveconfig
```

G i t ng l nh, c ph n h†i ‡ng l nh tr c khi ti p t c. V d th nh c ng: `$command,SAVECONFIG,response: OK` v i checksum n u receiver xu t. L~i ho c h t th•i gian ch• c n ”c x l•, kh ng g i reset l p l i. FRESET x…a c u h nh receiver; ch› dŠng trong quy tr nh kh i ph c c… ch‹ ch.

Unicore nh n l nh ASCII; kh ng g i g…i UBX c‹a u-blox. Backend c… th sinh log COM3 cho h d'y kh c; ESP32 nh tuy n c c l nh output RTCM/NMEA sang c•ng n i c‹a n…, kh ng •i t a hay th•i gian kh o s t.

9. X l s‡ c

D u hi u	Ki„m tra v h ng x l
Kh ng th y log console	ƒ‡ng c•ng USB/driver, 115200 baud, ngu†n v c p truy n d li u.
UNKNOWN/INVALID, kh ng c… GGA	RX/TX u chšo, chung GND, ‡ng c•ng receiver, ‡ng baud; ki m tra tr€c ti p receiver khi c n.
MQTT ”c nh ng CORS ch a qu n l• ”c	Broker/topic theo serial, thi t b „ nh n/c p v license; ki m tra command/result thay v ch› nh n k t n i TCP.
WebSocket kh ng k t n i	ƒ‡ng host/port endpoint thi t b ; firmware hi n dŠng ws, kh ng ph i wss. MQTT v‰n l k nh i u khi n kh c.
C… NMEA nh ng RTCM b ng 0	ƒ‡ng ch base, survey „ xong, ‡ng output UART v b n tin RTCM „ b t.
C… RTCM UART nh ng caster kh ng nh n	ƒ‡ng source account/mountpoint/version NTRIP; server ”c b t, kh ng kh…a v ‹ license/provision.
C… hai k t n i cŠng t i kho n rover	X c minh client n o o, client n o thu c d ch v monitor/decode tr n caster; kh ng x…a t i kho n khi ch a x c nh.
Auto base kh ng FIX	T i kho n rover, mountpoint u v o c… d li u, GGA m i, c•ng RTCM input, anten/ che khu t v kho ng c ch tr m.
Auto base thi u m‰u	Ki m tra FIX c… duy tr , GGA c… m i v ‹ m‰u h”p l trong c a s• l y m‰u.
C u h nh GNSS b o l˜i	ƒ c l˜i Unicore response timeout/rejected, ‡ng baud/c•ng v ph n h†i $command; kh ng coi NO MATCHING FUNC l OK.
Kh , i ng l i v‰n dŠng c u h nh c–	ƒ i chi u service ang c p t CORS v i c u h nh ATG; ki m tra l u NVS v k t qu tri n khai d ch v .
T a l ch dŠ rover FIX	H quy chi u, cao, chi u cao anten, t a base v phšp chuy n VN2000; ki m tra tr n m c c l p.

Khi g i y u c u h˜ tr”: cung c p serial, phi n b n, th•i i m l˜i, phase/l• do workflow, lo i receiver, baud/c•ng n i, server n o l˜i v log „ che m t khŒu/token. Kh ng g i l i reset nhi u l n khi ch a x c nh l˜i.

10. Nghi m thu v b o tr

1. Ch p/ghi nh n ‡ng u d‘y, ngu†n, anten, SIM/APN v serial ”c c p.
2. T’t/b t ngu†n ki m tra thi t b n p ‡ng nh n d ng, d ch v , lo i GNSS v c u h nh base.
3. V i fixed/survey: i chi u t a base v b n tin tham chi u sau khi receiver •n nh.
4. V i auto base: ki m tra t ng b c chuy n rover, nh n c i ch nh, RTK_FIXED, s m‰u, chuy n base v ph c h†i output.
5. DŠng rover c l p k t n i ‡ng mountpoint, ki m tra tu•i c i ch nh, FIX v i m m c „ bi t.
6. Ki m tra ri ng server 1/2, lu n ph t/theo y u c u v kh…a/m , t ng server; kh…a kh ng ”c b fail-open b• qua.
7. Th m t m ng r†i ph c h†i; ki m tra k t qu th€c t , kh ng ch› Ÿn ho c nh„n online.
8. Sau khi di chuy n anten, thay receiver/firmware ho c thay h t a , th€c hi n l i ph n nghi m thu li n quan.

Kh ng c… thao t c o ph n c ng t€ ng n o ”c th€c hi n ch› b , i vi c c ho c xu t b n t i li u n y. B n hi n t i „ i chi u code; nghi m thu ngo i hi n tr •ng do •n v tri n khai ghi nh n ri ng.

T i li u tham chi u

- B l nh ASCII v ph n h†i m t , ‘y ”c i chi u tr€c ti p v i b t o l nh v parser trong firmware V2.3.5; khi thay receiver/firmware GNSS c n ki m tra t i li u c‹a ‡ng model tr c khi p d ng.